nanoExperiment No. 11

Title

Shell Programming Using Bash

Aim

The problem is to write and execute Bash shell scripts that demonstrate interactive input, positional parameters, arithmetic operations, conditional statements, and loop constructs, thereby automating basic computational and logical tasks.

Theory

Shell Programming

Shell scripting is a method of automating tasks by writing a sequence of Linux commands in a script file executed by the shell.

Arithmetic Operations

Arithmetic can be performed using:

$((expression))

Interactive Scripts

Interactive scripts accept input from the user during execution using the read command.

Positional Parameters

Shell scripts accept command-line arguments using:

- $0 – Script name
- $1, $2, ... – Positional parameters

Conditional Statements

Decision-making constructs include:

- if
- if-else
- elif

Loops

Loops execute commands repeatedly:

- for
- while

COMMANDS USED-

read	Accept user input
echo	Display output
$(( ))	Arithmetic operations
if, elif, else	Conditional statements
for, while	Loop constructs
chmod	Make script executable

Procedure / Steps

A. Interactive Shell Script

1. Create a script file: nano interactive.sh
2. Enter the following code:
   #!/bin/bash
   echo "Enter your name:"
   read name
   echo "Welcome $name"
3. Save and exit.
4. Make the script executable:
   chmod +x interactive.sh
5. Execute the script:
   ./interactive.sh

```
ubuntu@seema-ubuntu: ~
ubuntu@seema-ubuntu:~$ nano interactive.sh
ubuntu@seema-ubuntu:~$ chmod +x interactive.sh
ubuntu@seema-ubuntu:~$ ./interactive.sh
enter your name
seema
welcome seema
ubuntu@seema-ubuntu:~$
```

B. Using Positional Parameters

Create another script:
nano params.sh
Enter the following code:
#!/bin/bash
echo "Script Name: $0"
echo "First Argument: $1"
echo "Second Argument: $2"
Execute: ./params.sh Linux Bash

nano para

```
ubuntu@seema-ubuntu: ~
  GNU nano 7.2                                    params.sh *
#!/bin/bash
echo "script name:$o"
echo "first argument :$1"
echo "second argument :$2"
```

```
ubuntu@seema-ubuntu:~$ nano params.sh
ubuntu@seema-ubuntu:~$ ./params.sh linux bash
bash: ./params.sh: Permission denied
ubuntu@seema-ubuntu:~$ ./params.sh Linux Bash
bash: ./params.sh: Permission denied
ubuntu@seema-ubuntu:~$ chmod +x params.sh
ubuntu@seema-ubuntu:~$ ./params.sh Linux Bash
script name:
first argument :Linux
second argument :Bash
ubuntu@seema-ubuntu:~$
```

C. Arithmetic Operations

Create a new script file:

nano arithmetic.sh

Enter the following script:

```
#!/bin/bash
a=10
b=5
sum=$((a + b))
echo "Sum = $sum"
```

D. Conditional Statement

Create script:

nano condition.sh

Enter: #!/bin/bash

```
echo "Enter a number:"
read num
if [ $num -gt 0 ]
then
echo "Positive number"
else
echo "Zero or Negative number"
fi
```

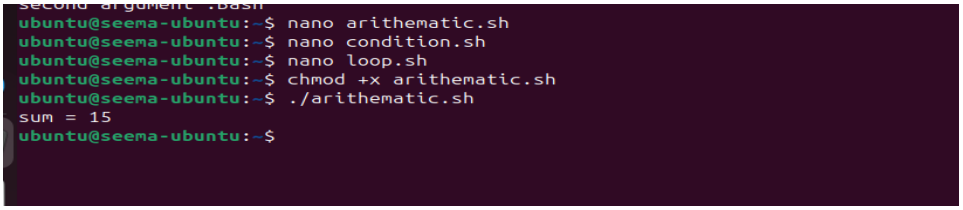
E. Loop Constructs

Create for-loop script:

nano loop.sh

Enter:

```
#!/bin/bash
for i in 1 2 3 4 5
do
echo "Iteration $i"
done
```



Outcomes

- ✓ Scripts accept user input and display results correctly.
- ✓ Positional parameters are accessed successfully.
- ✓ Arithmetic operations produce correct output.
- ✓ Conditional statements and loops execute as expected.

Result

- ❖ Bash shell scripts demonstrating interactive input, positional parameters, arithmetic operations, conditional statements, and loops were successfully created and executed.